

When the Guardrail Was Never Removed: A Controlled Hallucination Stress-Test of a Domain-Specific RAG Chatbot

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Abstract—This paper documents the construction and evaluation of a domain-specific Retrieval-Augmented Generation (RAG) chatbot over a constructed municipal tourism corpus of three plain-text documents. The pipeline chunks the corpus into 37 passages (500 characters, 50 overlap), indexes all-MiniLM-L6-v2 embeddings in an in-memory ChromaDB, retrieves the three nearest passages per query, and generates answers with openai/gpt-oss-20b via Groq. A controlled stress-test raises temperature from 0 to 1.0 and deletes the prescribed refusal clause while holding all else constant. Contrary to expectation, the ungrounded configuration refused all three out-of-domain samples and introduced no unsupported claims. Analysis attributes this to an unremoved grounding constraint conflated with the refusal clause, and proposes an ablation to separate them.

Keywords—retrieval-augmented generation, hallucination mitigation, vector database, prompt engineering, negative result

I. SYSTEM ARCHITECTURE

A. Corpus and Document Loading

Three plain-text files were placed in `./my_data/`: a city profile (history, festivals, landmarks, transport; 4,873 characters), a twelve-item visitor FAQ (office hours, tour requests, permits; 3,746) and a heritage guidelines document (classification scheme, Heritage Council rules, nomination and conduct; 4,336), totalling 12,955 characters. They are a constructed corpus authored for this laboratory, modeled on public municipal tourism material and not official city publications, which gives exact ground truth for every answer. `DirectoryLoader` converts each file into a `Document` whose `metadata.source` records its path. The handout’s default unstructured parser silently returned zero documents for these `.txt` files; `loader_cls=TextLoader` resolved it.

B. Chunking

`RecursiveCharacterTextSplitter` divides each document into passages of at most 500 characters with 50 of overlap, cutting on paragraph, line, sentence or word boundaries in that priority; it yielded 37 chunks averaging 355 characters. Chunking is required because an embedding model emits one fixed-length vector per input: encoding a whole document would average unrelated facts into one vector and destroy retrieval precision. Overlap ensures a fact spanning a boundary survives in at least one chunk.

C. Embedding and Indexing

Each chunk is encoded by all-MiniLM-L6-v2, a six-layer sentence-transformer producing 384-dimensional vectors [2, 3]; it runs locally, so no text leaves the runtime. Vectors, text and

metadata are written into an in-memory ChromaDB collection.

D. Retrieval and Generation

The store is exposed as a retriever with $k = 3$. Nearest-neighbour search has no relevance floor, so an out-of-domain query still yields three passages, merely irrelevant ones; correct refusal is therefore decided downstream by the generator, not by empty retrieval. `create_stuff_documents_chain` concatenates the chunks into the `{context}` placeholder, and `create_retrieval_chain` binds retriever and generator into one runnable that returns the answer with its retrieved context, so every claim traces to its source passage [1, 5]. The handout’s `llama-3.1-8b-instant` was deprecated by Groq on 17 June 2026 and decommissioned on 16 August 2026 (HTTP 404); Groq’s designated replacement, `openai/gpt-oss-20b`, was substituted and used identically in both configurations [4].

II. IMPLEMENTATION DETAILS

Table III lists every configured parameter; all were held constant across configurations except temperature and the fallback rule.

The API credential is supplied at runtime through `getpass.getpass()` and held only in `os.environ`, never in the notebook. The handout’s `not in os.environ` guard was replaced with a clear-then-prompt, since the guard makes a stale key uncorrectable and yields an unexplained HTTP 401.

III. EXPERIMENTAL RESULTS AND HALLUCINATION ANALYSIS

A. Experimental Design

Configuration A (baseline) uses temperature 0 and a system prompt with an explicit refusal clause: reply “I cannot answer based on the provided domain data” when the context is insufficient. Configuration B raises temperature to 1.0 and deletes that clause; everything else is identical. Three in-domain queries established baseline correctness; one out-of-domain query (Earth–Mars distance and Apollo 11 propulsion, absent from all files) was posed to both configurations, sampled three times under B because decoding there is stochastic, and one in-domain control query was posed to both.

B. Results and Analysis

Table I lists every query with expected and actual output; Table II compares the configurations side by side. The baseline behaved as designed: the prescribed refusal was returned verbatim on the out-of-domain query and every in-domain query was answered correctly, each claim traceable to a printed chunk. Retrieval was not empty in the out-of-domain case (two FAQ chunks, one guidelines chunk), so the refusal was a judgement

TABLE I
TEST QUERIES, EXPECTED RESPONSES, AND MODEL OUTPUTS

#	Test Query	Type / Cfg	Expected Response	Actual Model Output
1	SUMAKAH acronym meaning	In / A	SUMAsayaw, MASayahin, KAlakalan, Hardin	SUMAsayaw – dancing; MASayahin – cheerful; KAlakalan – commerce; Hardin – garden
2	Tour lead time and max group size	In / A	10 working days; 40 participants	At least ten (10) working days; maximum 40
3	Heritage Council size and quorum	In / A	7 members; quorum of 4	Seven (7) members; quorum of four (4)
4	Earth–Mars distance; Apollo 11 propulsion	Out / A	I cannot answer based on the provided domain data.	I cannot answer based on the provided domain data. (verbatim)
5–7	Same as #4, runs 1–3	Out / B	No prescribed refusal available	(1) “I’m sorry, but I don’t have that information in the context provided.” (2) “I’m sorry, but I don’t have that information available in the provided context.” (3) identical to run 1. No fabricated content in any run.
8	Tour lead time and max group size (control)	In / B	10 working days; 40 participants	At least 10 working days; maximum 40

TABLE II
BASELINE VERSUS UNGROUNDED CONFIGURATION

Criterion	A ($t=0$, fallback ON)	B ($t=1.0$, fallback OFF)
Refusals on out-of-domain input	1 / 1	3 / 3
Prescribed fallback string used	Yes (verbatim)	No — self-worded in all runs
Determinism across runs	Deterministic (greedy)	Runs 1 and 3 identical; run 2 differs by two words
Unsupported claims introduced	0	0
In-domain accuracy	4 / 4	1 / 1 (control)

about the sufficiency of the context, not a consequence of retrieving nothing.

The ungrounded configuration did not hallucinate. It refused in all three samples, never in the prescribed wording, and every refusal explicitly attributed the gap to the provided context (Table I, rows 5–7). No Earth–Mars figure and no Apollo 11 description appeared in any output, although both sit within the model’s pretraining knowledge: refusal rate 4/4, hallucination rate 0/4. Determinism barely degraded at all: runs 1 and 3 were byte-identical and run 2 differed only by the insertion of “available,” so sampling at temperature 1.0 varied the surface wording of the refusal without bringing any fabricated continuation close to competitive. The in-domain control was answered correctly by both configurations. Correctness and grounding remain different properties [6]: an accurate Earth–Mars figure from B would still have breached the grounding contract, since it appears in no retrieved chunk.

The negative result is explained by the design, not the execution. The prescribed modification deletes the sentence specifying the verbatim refusal string but leaves intact the preceding constraint, “Answer questions strictly using ONLY the provided context below.” That line alone kept the model grounded; what was removed was the *output format* of the refusal, which is ex-

actly what the results show; the two lines are not independent variables. Temperature had the smaller effect, yielding differently worded refusals rather than different answers. Model substitution also limits generalisation: `openai/gpt-oss-20b` is newer and larger than the 8B Llama the exercise targeted, so the result is model-specific. A four-condition ablation (both lines; refusal clause only removed; grounding constraint only removed; both removed), sampled at each temperature, would separate “the refusal clause was unnecessary” from “the grounding constraint did all the work.”

IV. CONCLUSION

Building this chatbot made concrete a lesson that is easy to underestimate: retrieval alone does not produce a grounded system. The retriever returned three passages even for a spaceflight question posed to a tourism corpus, and nothing in the retrieval layer signalled that they were irrelevant; responsible behaviour was decided entirely by the generator’s instructions. Prompt restrictions proved more load-bearing than the experiment was designed to detect: removing the model’s sanctioned way to decline did not stop it declining, because a second constraint I had not identified as separate was still present—a flaw a confirming result would have hidden. Retrieval-based information contributes auditability: the chain returns its chunks with the answer, so a claim traceable to no chunk is a hallucination by definition, whether or not it is true. Finally, supplying the API key through runtime prompting rather than a hard-coded literal is a small habit with an asymmetric payoff, since a key embedded in a shared notebook is surrendered to every future reader. Together, prompt constraints, traceable retrieval and credential discipline separate a demonstration that happens to work from a system that can be trusted on inputs its authors never anticipated.

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TABLE III
IMPLEMENTATION PARAMETERS

Parameter	Value
Document loader	DirectoryLoader + TextLoader (unstructured returned 0 docs)
Chunk size / overlap	500 / 50 characters (10%)
Realised chunks	37 (avg. 355 chars) from 3 files, 12,955 chars
Embedding model	all-MiniLM-L6-v2 (384-d, local)
Vector store	ChromaDB (in-memory, session-scoped)
Retrieval top- k	3
Generation model	openai/gpt-oss-20b via Groq (replaces llama-3.1-8b-instant)
Temperature (Cfg A / B)	0 / 1.0
Fallback rule (Cfg A / B)	Enabled / Removed (independent variable)
Combination strategy	stuff (~1,065 chars of context)